become lost in the abstract nature of the complex algebra and loose all connection to the real world. Your task is to merge the two views together. The Laplace transform is the primary method for analyzing electric circuits. Keep in mind that *any* system governed by differential equations can be handled the same way; electric circuits are just an example we are using.

The brute force approach is to solve the differential equations controlling the system, providing the system's impulse response. The impulse response can then be converted into the s-domain via Eq. 32-1. Fortunately, there is a better way: transform each of the individual components into the s-domain, and *then* account for how they interact. This is very similar to the *phasor transform* presented in Chapter 30, where resistors, inductors and capacitors are represented by R, $j\omega L$, and $1/j\omega C$, respectively. In the Laplace transform, resistors, inductors and capacitors become the complex variables: R, sL, and $1/sC$. Notice that the phasor transform is a *subset* of the Laplace transform. That is, when σ is set to zero in $s = \sigma + j\omega$, R becomes R, sL becomes $j\omega L$, and $1/sC$ becomes $1/j\omega C$.

Just as in Chapter 30, we will treat each of the three components as an individual system, with the current waveform being the input signal, and the voltage waveform being the output signal. When we say that resistors, inductors and capacitors become R, sL , and $1/sC$ in the s-domain, this refers to the output divided by the input. In other words, the Laplace transform of the *voltage waveform* divided by the Laplace transform of the *current waveform* is equal to these expressions.

As an example of this, imagine we force the current through an inductor to be a unity amplitude cosine wave with a frequency given by ω_0. The resulting *voltage* waveform across the inductor can be found by solving the differential equation that governs its operation:

$$v(t) = L\frac{d}{dt}\,i(t) = L\frac{d}{dt}\cos(\omega_0 t) = \omega_0 L\,\sin(\omega_0 t)$$

If we start the current waveform at $t = 0$, the voltage waveform will also start at this same time (i.e., $i(t) = 0$ and $v(t) = 0$ for $t < 0$). These voltage and current waveforms are converted into the s-domain by Eq. 32-1:

$$I(s) = \int_0^\infty \cos(\omega_0 t)\, e^{-st}\, dt = \frac{\omega_0}{\omega_0^2 + s^2}$$

$$V(s) = \int_0^\infty \omega_0 L \sin(\omega_0 t)\, e^{-st} dt = \frac{\omega_0 L s}{\omega_0^2 + s^2}$$